KIIT University

ExcelR Edtech · SAP Industrial Elective - BTP Developer

CAPSTONE PROJECT REPORT

SAP BTP Integration Suite

Employee Data Integration for Hire-to-Retire (H2R) Process

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1. Problem Statement

In modern enterprises, HR systems must share employee data seamlessly across payroll engines, ERP platforms, and self-service portals. Manual transfers introduce inconsistencies, delays, and compliance risks. The Hire-to-Retire (H2R) process - spanning onboarding, role changes, transfers, and exit - demands real-time, automated synchronisation at every stage. Without robust integration middleware, organisations face data silos and operational bottlenecks that directly impact workforce management and downstream system accuracy.

2. Proposed Solution

This project implements a complete end-to-end SAP BTP Integration Suite scenario simulating an employee onboarding data flow from an HR source system to downstream enterprise platforms. The iFlow Employee_Data_Fetch exposes a secure HTTPS endpoint, fetches live employee data from an external REST API, applies field-level transformation via Content Modifier and Groovy Script, and returns a structured JSON payload - with a built-in Exception Subprocess handling failures gracefully. The architecture mirrors real SAP BTP to SuccessFactors and S/4HANA enterprise integration patterns.

3. Key Features and Architecture

Feature	Description
HTTPS Trigger	Secure inbound endpoint (/employeedata) accepts integration requests from upstream systems
Real-time Data Fetch	HTTP GET to JSONPlaceholder REST API returns live structured employee JSON payload
Data Transformation	Content Modifier structures payload; Groovy Script extracts key fields into clean output JSON
Exception Handling	Exception Subprocess returns structured error JSON with H2R-Onboarding process context on failure
Cloud-Native Deployment	Runs entirely on SAP BTP Cloud Foundry - no on-premise infrastructure required
Request-Reply Pattern	Industry-standard synchronous SAP integration pattern with guaranteed API response

4. Technology Stack

Technology	Version	Role in Project
SAP BTP	Trial	Cloud platform - all services, subaccounts, and runtime environments
SAP Integration Suite	Trial Plan	Integration middleware - iFlow design, deployment, and runtime
Cloud Integration	Trial Tenant	Design, configure, deploy, and monitor integration flows
HTTPS / HTTP Adapters	Built-in	Inbound HTTPS trigger; outbound HTTP GET to external REST API
Content Modifier	Built-in	Structures and enriches the employee data payload mid-flow
Groovy Script	Built-in	Field-level data extraction and clean JSON output formatting
JSONPlaceholder API	Public REST	External employee data source - GET /users/1
SAP BTP Cockpit	Web Console	Subaccount management, role assignment, service subscriptions
SAP Business App Studio	Trial	Cloud IDE for BTP development and iFlow configuration

5. End-to-End Flow Explanation

Step 1 - Trigger: An HR system sends an HTTPS request to /employeedata on SAP BTP Cloud Foundry, simulating an H2R onboarding event.

Step 2 - Fetch: Fetch_Employee_API (Request Reply) calls JSONPlaceholder via HTTP Adapter; receives raw employee JSON with name, email, address, and company fields.

Step 3 - Transform: Content Modifier 2 structures the payload. Groovy Script (Transform_Employee) extracts required fields and formats a clean, standardised JSON aligned to downstream expectations.

Step 4 - Response: End Event returns the transformed, validated JSON to the calling system, completing the synchronous integration cycle.

Step 5 - Error: If any step fails, Exception Subprocess fires: Content Modifier returns { "status":"ERROR", "message":"Employee data fetch failed", "process":"H2R-Onboarding", "handler":"SAP-BTP-Exception-Subprocess" }.

6. Implementation Screenshots

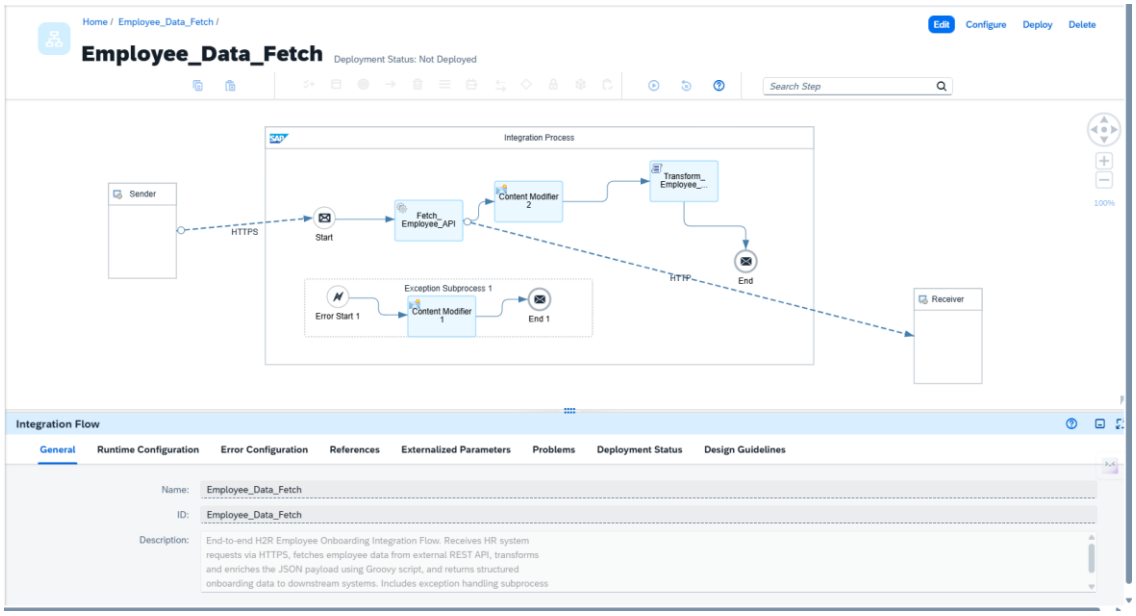

Figure 1: Employee_Data_Fetch - Sender (HTTPS) → Fetch_Employee_API → Content Modifier 2 → Transform_Employee → End, with Exception Subprocess

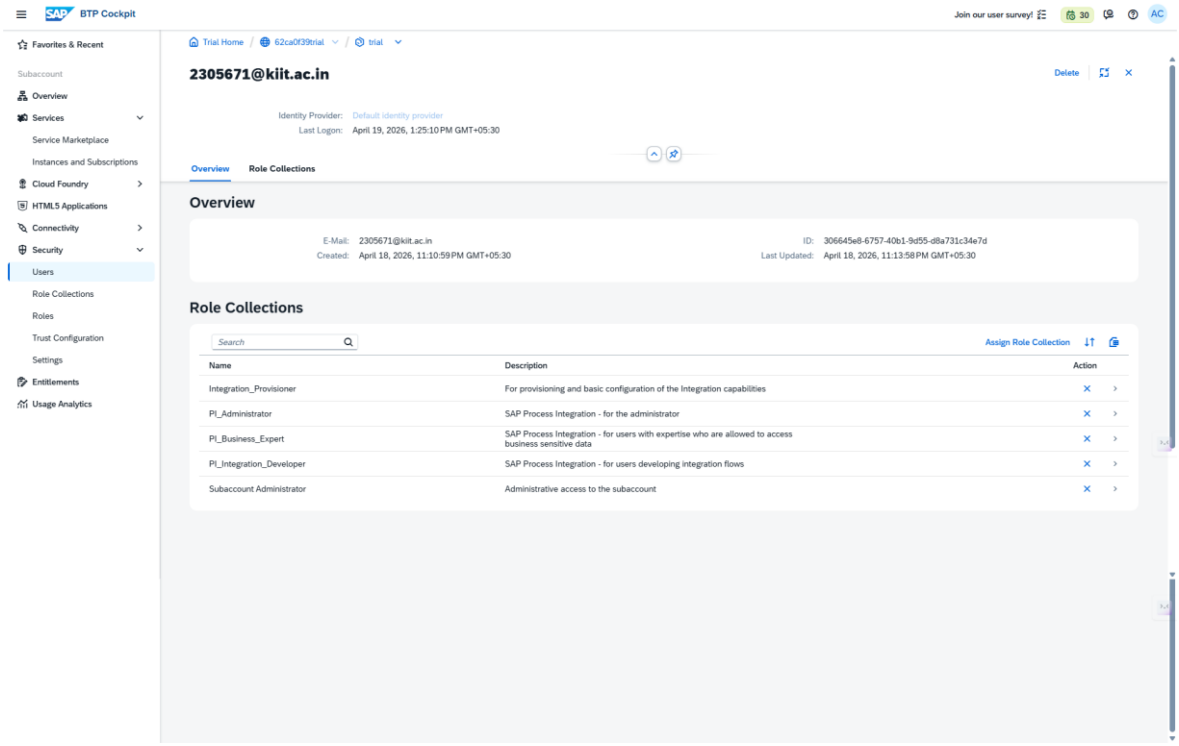

Figure 2: All required roles assigned to 2305671@kiit.ac.in - Integration_Provisioner, PI_Administrator, PI_Business_Expert, PI_Integration_Developer

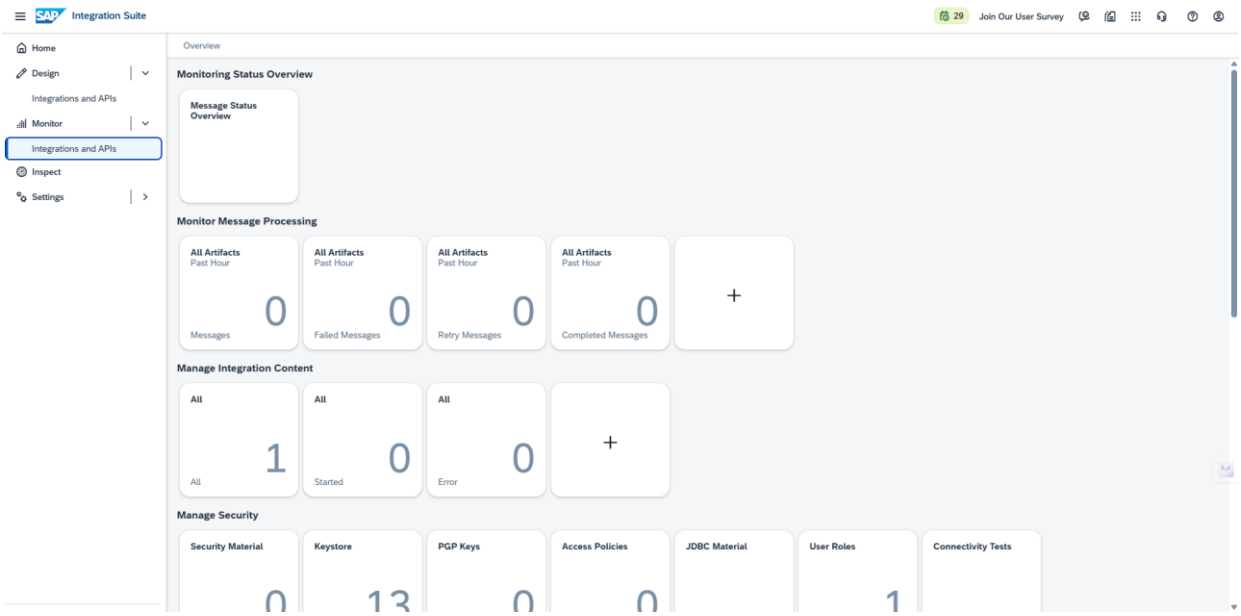

Figure 3: Roles assigned to 2305671@kiit.ac.in; Monitor showing iFlow deployed Apr 19 2026; BTP Cockpit with Integration Suite and SAP BAS subscribed

7. Unique Points

Cloud-Native, Zero On-Premise: Entire solution on SAP BTP Cloud Foundry - no local SAP install, instantly scalable, fully managed.

Full Integration Stack: Implements all enterprise layers in one iFlow - HTTPS trigger, Request-Reply fetch, Content Modifier and Groovy Script transformation, Exception Subprocess error handling.

Built-in Resilience: Structured error JSON with explicit H2R-Onboarding process context enables graceful downstream failure handling without manual intervention.

Production-Ready Pattern: Request-Reply + HTTPS/HTTP is the exact pattern used in live SAP BTP to S/4HANA and SuccessFactors production integrations.

Extensible Without Redesign: OAuth 2.0, S/4HANA OData, and audit logging can be added as independent modular components - no core iFlow changes needed.

8. Future Improvements

#	Improvement	Description
1	OAuth 2.0 Security	Secure HTTPS endpoint with client credentials for production-grade access control
2	S/4HANA OData Connection	Replace mock API with real SAP S/4HANA HR OData service for live enterprise data
3	SuccessFactors Connect	Connect to SAP SuccessFactors Employee Central for actual H2R lifecycle events
4	Audit Logging	Trace-level logging and monitoring dashboards for compliance and visibility
5	Bulk / Batch Processing	Paginated API calls to synchronise full employee datasets in batch mode
6	Content-Based Routing	Route data to different targets based on department, role, or employment type

9. Conclusion

This capstone project delivers a production-aligned SAP BTP Integration Suite implementation for the Hire-to-Retire (H2R) process. The Employee_Data_Fetch iFlow demonstrates end-to-end integration competency - secure HTTPS trigger, live REST API fetch via Request-Reply, field-level transformation using Content Modifier and Groovy Script, and resilient error handling via a structured Exception Subprocess - all deployed on SAP BTP Cloud Foundry with proper role-based access control. The architecture reflects real enterprise SAP BTP integration patterns and provides an extensible foundation for production scenarios involving SAP S/4HANA, SuccessFactors, and enterprise-grade security.

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